Wazuh SIEM SOC Home Lab

Overview

Designed and configured a virtual Security Operations Center (SOC) using Wazuh SIEM, Windows 10, Windows Server, Ubuntu Linux, Kali Linux, and Sysmon.

Project Objectives

- Centralize Windows security event monitoring.
- Detect failed login attempts and administrative privilege changes.
- Develop custom detection rules.
- Investigate suspicious activity using Wazuh alerts.
- Test incident response configurations in a controlled environment.

Lab Environment

System	IP Address	Purpose
Windows 10	10.10.10.10	Endpoint monitoring and Sysmon
Windows Server	10.10.10.20	Windows security events
Ubuntu Wazuh	10.10.10.30	SIEM manager and dashboard
Kali Linux	10.10.10.40	Authorized security testing

Technologies

Wazuh 4.14.8, Sysmon, Windows Event Viewer, PowerShell, Ubuntu, Kali Linux, VirtualBox, XML.

Detection Engineering

- Rule 100100: Windows failed login (Event ID 4625).
- Rule 100104: User added to local Administrators group (Event ID 4732).
- Rule 100110: Multiple failed login attempts within 120 seconds.

Results

- Successfully connected and monitored Windows endpoints.
- Verified custom detection alerts in Wazuh.
- Investigated authentication and administrative group changes.
- Configured active-response components for controlled testing.

Security Note

This repository contains sanitized educational examples. No production credentials or sensitive records are included.